

ESPERANZA II

*Lunar-Martian Hybrid Regolith Intelligence System
for Deep Space Food Production*

NASA Mars to Table Challenge 2026 — Solution Summary

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Prior Recognition: NASA LunaRecycle Challenge Winner (ESPERANZA I)

Team	Challenge	Deliverable	Deadline
ESPERANZA Research	Mars to Table 2026	Solution Summary	August 14, 2026
EXECUTIVE ABSTRACT — ESPERANZA II is the first integrated food production system designed to simultaneously resolve the three critical threats to Mars surface agriculture — perchlorate toxicity, mineral sterility, and radiation-induced gut microbiome destruction. It combines real NASA geochemical data from 82 verified regolith samples (35 Mars + 47 lunar) plus 400 ISS microbiome records with a Darwinian Genetic Algorithm (DGA), a Microbial Community-scale Metabolic Model (MCMM) adapted for Mars, and a closed-loop biometric controller (NEXUS-Loop) that represents a world first in space food system design. The system produces a complete and nourishing diet for 15 astronauts across a 500-sol Mars surface mission (513.75 Earth days) without Earth resupply, within 81.2 of the 90 food-specialist hours available per 5-sol week — crew time being the only rationed resource. The NEXUS-Loop — the fourth innovation — closes the biometric→microbiome→meal cycle within a single sol: a non-invasive wristband drives MCMM-based SCFA forecasts that re-optimize the DGA formula, grounded in the author's open analysis (preprint) of NASA OSDR Inspiration4 flight data (n = 4, LEO) (p = 0.026; DOI 10.21203/rs.3.rs-10232328/v1). Direct Earth applications: perchlorate bioremediation (>500M people in arid regions), gut-health nutrition for IBD (~10M patients) and cyanobacterial biofertilisation — detailed in §6.			

1. THE PROBLEM: WHY MARS CANNOT FEED ITS CREW

A 500-sol crewed Mars surface mission (513.75 Earth days) faces three simultaneous threats that no current food system architecture resolves — what ESPERANZA II terms the Triple-Threat Syndrome. Each is individually fatal to conventional agriculture.

(1) Perchlorate toxicity (ClO_4^- 0.4–0.6 wt% at the Phoenix site, Hecht 2009; perchlorate has since been identified at widely separated landing sites) — thyroid-disrupting and crop-lethal, no closed-loop bioremediation in any prior architecture. (2) Mineral sterility — Mars regolith has $\text{FeO}/\text{CaO}/\text{MgO}$ but <5% bioavailability due to alkaline pH 7.8–8.5 and absent organic ligands; a 100% Mars-only substrate yields no edible crop. (3) Radiation-induced gut dysbiosis — ISS, MARS500 520-day analog and Twins Study confirm chronic SCFA collapse and inflammatory phenotype no current food system addresses.

2. THE ESPERANZA II INNOVATION: FOUR WORLD FIRSTS

World First 1 — Lunar regolith as targeted mineral corrector. 47 verified Apollo+LRO samples vs. 35 Mars samples reveal precise geochemical complementarity: lunar highlands provide bioavailable Ca/Mg (anorthite weathering, pH 6.8) that Mars regolith lacks. Optimal blend 32.1% lunar + 21.3% Mars achieves 4.2% Fe and 5.8% Ca bioavailability vs. <0.5% in Martian regolith alone. Lunar regolith flies as GCR shielding on the ~7-

month transit and is then repurposed as an Fe/Ca corrector — multi-mission resource efficiency. If the lunar stockpile is unavailable at departure, Contingency Mode shifts HybridSoil to 100% Mars-derived substrate (LunaRex-treated regolith + recycled crew biomass); the ~15% Fe/Ca shortfall is met by Earth-supplied mineral supplements, which carry negligible calories: the by-calorie fraction is unchanged at 47.0%, under the ≤50% cap.

World First 2 — Darwinian Genetic Algorithm (DGA) for probiotic optimisation. Trained on 400 ISS microbiome records (Voorhies 2019, Turrone 2020). 4,000 candidate formulas evolve across 5 fitness functions (radiation shielding, cosmic-ray repair, gut barrier, mood, immunity). Fitness 10.15/11.0 (92.2%, 5-fold CV ±0.42, 95% CI [9.73-10.57]).

World First 3 — Every meal as functional radiation medicine. Space food has never embedded non-nutrient bioactives (Heliyon 2024 PMC11466544). ESPERANZA II ferments living probiotic biomass into 78 meals (kefir, miso, tempeh) so radiation-protective metabolites dose continuously, not as supplement.

World First 4 — NEXUS-Loop: real-time biometric → microbiome → meal closed loop. A non-invasive wristband (HRV) feeds an MCMM (Quinn-Bohmann 2024) adapted for Mars; the SCFA forecast triggers per-sol DGA re-optimisation, with salivary cortisol confirming a flagged state rather than screening routinely. Demonstrated, not asserted: on NASA OSDR Inspiration4 data (n = 4, short-duration LEO analogue), fusing all modalities separated pre/post-flight state at LOSO 0.708 against a 0.489 ± 0.107 null (p = 0.026), while no single biomarker survived correction (0 of 1112). This is a method claim, not a wearable performance claim: NEXUS is a fused state detector, never a single-marker screen (Puerta Angulo 2026, preprint, DOI 10.21203/rs.3.rs-10232328/v1; Appendix 3.2 §9).

3. SYSTEM ARCHITECTURE: FIVE INTEGRATED MODULES

ESPERANZA II is built as five coupled modules, each parameterised in the system model (ConOps Appendix A8): (1) LunaRex - Random Forest + Gaussian-process lunar mineral mapping (77% accuracy); (2) MarsRem - Monod-kinetic ODE for Azospira-driven perchlorate bioremediation, reaching 0.017 wt% ClO₄⁻ by Sol 24 (criterion: EPA RfD 0.7 ug/kg-day, IRIS/NRC 2005 [15]; crop tissue is a HACCP CCP); (3) HybridSoil - Dirichlet-Pareto 6-objective optimiser (32.1% lunar + 21.3% Mars + 20% cyanobacterial + 24.4% organic + 2.2% binder); (4) CyanoGrow - Arthrospira + Anabaena N₂-fixation at 100 hPa pN₂ (Verseux 2021); (5) FermentLab + NEXUS - DGA fermentation coupled to MCMM-SCFA prediction and per-astronaut probiotic adjustment.

Bio-Cycle Recovery (FermentLab sub-process): all inedible crop biomass — stems, roots, husks — is channelled into Tenebrio molitor rearing and fungal fermentation, achieving 100% internal re-assimilation with zero agricultural waste exported to the primary habitat ECLSS (satisfies Circular Resource Systems sub-criterion, Rules Appendix C p.18).



Figure 1 — Mass and information flow across the five ESPERANZA II modules and the NEXUS closed-loop AI layer. Closed loops with ECLSS for O₂, N₂, water and CO₂.

4. ECLSS INTEGRATION AND CLOSED-LOOP RESOURCE MANAGEMENT

ESPERANZA II is architecturally designed for full integration with the Environmental Control and Life Support System (ECLSS) across three resource cycles, ensuring near-closed-loop operation compliant with the Challenge requirements.

Oxygen: crops release 9.82 kg O₂/sol, 76% of crew demand (12.95 kg/sol; Anderson 2018). MarsRem is claimed as a detoxification process only: its chlorite-dismutase step generates O₂ intracellularly, which the organism respires in situ (Melnyk 2011), so no MarsRem oxygen is credited to the atmospheric balance. Carbon: crops (13.5) and spirulina (7.2) together fix 20.7 kg CO₂/sol. Crew respiration supplies 15.4 kg/sol of

that — 74%, the primary carbon feed — with fermentation, composting and heterotrophs returning a further 2.8; the remaining 2.5 kg/sol is drawn from abundant Martian CO₂ (3.1 kg/sol makeup, 0.6 buffered). Using in-situ CO₂ to feed photosynthesis is a design choice, not a shortfall (ledger in ConOps Appendix A3). Water: CyanoGrow hydroponic transpiration recaptured by condensers; FermentLab water reclaimed; net consumption <0.8 L/astronaut/day. Nitrogen: crew urine + composted feces form the 24.4% organic fraction of HybridSoil; Anabaena fixes N₂ from the 1.9% N₂ Martian atmosphere (Verseux 2021). Earth-provisioned fraction: 47.0% by calories (Rules p.17, Table 3; cap 50%) — 41.2% by per-ingredient USDA audit, plus a 5.8% declared reserve for menu draw above the crop-table nameplate.

Habitat envelope: 8.2 psi (56.5 kPa), 34% O₂ hyperoxia, 18–27 °C, 3.71 m/s². Fermentation/crop bays in fire-rated enclosures with O₂ trim per NASA-STD-3001 Vol.2 Sec. 6.4. ECLSS coupling is bidirectional: ESPERANZA II receives the crew's full 15.4 kg/sol CO₂ + 12 L/sol grey water + 644 L/sol canopy transpiration recaptured at 98.5% (9.7 L/sol net makeup); returns 15.02 kg/sol O₂ from crops and spirulina (116% of crew demand), N₂ (CyanoGrow biofixation) and 98.5% recovered water. Lighting is LED sole-source (3.0 μmol/J): Σ(DLI×area) over 373 m² canopy = 30.4 kW mean / 33.3 kW peak — wheat runs 24/0 as baseload while soybean and potato tessellate opposite 12.33 h windows, holding peak within 9% of mean. Food system: 58.1 kW mean / 64.0 kW peak / 77 kW installed; ~300 m² dust-derated radiator at 40 °C. Crop chambers hold their own atmosphere (21% O₂, 1,200–1,500 ppm CO₂) behind an airlock — at 34% O₂ these C3 crops lose 15–25% of net photosynthesis to photorespiration; LED drivers sit in a normoxic bay (NASA-STD-6001). Footprint: 511 m² growing area (373 canopy + 40 spirulina + 48 Tenebrio + 50 mushroom) within 1,058 m³ of pressurised food-system volume (per-module breakdown in the Design Layout).

Module	TRL (current)	Mass (kg)	Power kW (kWh/sol)
LunaRex (regolith mapping ML)	3	120	0.6 (14.8)
MarsRem (perchlorate bioremediation)	4	480	2.1 (51.8)
HybridSoil (Pareto-optimised substrate)	4	210	0.4 (9.9)
CyanoGrow (PBR cyanobacteria + N ₂ fix)	5	780	4.8 (118.4)
FermentLab (probiotic + Tenebrio)	5	410	1.9 (46.9)
NEXUS (closed-loop biometric AI)	3–4	85	0.9 (22.2)
Total ESPERANZA II	3–5	2,085	10.7 (263.9)

Earth-provisioned budget (47.0% by calories, NASA cap ≤50%, Rules p.17 Table 3): 25,729 kg launch: biological starters (seeds, cultures, colonies) 205 kg, micronutrient premix 180 kg, allergen-free emergency rations 1,500 kg (90 sols), Earth-provisioned food 6,274 kg, system hardware, spares & consumables 17,570 kg. 53.0% of calories are in-situ; no resupply assumed.

5. OPERATIONAL PROTOCOLS: FOUR MISSION POINTS

MISSION POINT	OPERATIONAL DETAIL
Startup (Sols 1–60)	Overlaps outgoing crew (their Sols 450-500): the system is inherited in production - fresh output is continuous from Sol 1, with no cold start and no production gap. Control transfers progressively: Sol 1 NEXUS handover + water → Sol 7 regolith modules under incoming control → Sol 24 ClO ₄ ⁻ <0.017 wt% re-verified → Sol 30 full authority transfer + HACCP re-validation (10 CCPs) → Sol 35 first harvest of incoming-crew planting. Sol-60 gate: ≥53% in-situ kcal, ≥35% of it from the incoming crew's own plantings; water recycling ≥95%.

Nominal (Sols 60–450)	24,141 kcal/sol in-situ (53.0% of 45,525), the crop-table nameplate; the 2,628 kcal/sol drawn above it is booked to Earth stock. 14-sol rotation, 12 sources. NEXUS: 200+ sensors, per-sol DGA re-opt. 98.5% water / 98.3% organics recycled. 2 specialists @ 45 h/5-sol; 13 crew on cooking rota; 4 cross-trained backups. NEXUS ladder: NOMINAL → WATCH (HRV <38 ms, fused index) → confirmatory salivary cortisol → ALERT (HRV <28 ms / cortisol >18 nmol/L / gut <42) → CRITICAL (gut <30 + medical).
Contingency (any sol)	Yellow (single crop, ≤14 sols) → Orange (module-wide, 30–50% ration supplement) → Red (total failure, 100% rations + decontamination). Reserve: 4.10 M kcal = 90 sols full / 180 sols at 50%. NEXUS load-shedding; 4 backups activate; quarantine + peracetic acid/UV-C, 3 negatives before restart, Earth notification <24 h.
Shutdown (Sols 450–500)	Production sustained + incoming-crew prep: seed stock + germination; culture bank check (kefir, koji, miso, spirulina, Tenebrio, mushroom); equipment audit; docs update. Crop stagger aligned to next crew's Sol 1–30 handover. 50-sol formal transfer with 2 incoming cross-trained. Success: 100% viable cultures, zero production gap, HACCP + NEXUS logs archived.

Monitoring cadence — per sol: NEXUS health-metric scan, perchlorate residual, pH, fermentation Aw (NEXUS auto + nutrition specialist sign-off). Per 5-sol week: crop yield vs. forecast, Tenebrio biomass, HRV/cortisol per crew (food systems engineer). Per 14-sol cycle: meal-plan rotation closure, DGA re-fitness, HACCP audit (both specialists + commander). Hand-over: 50-sol overlap, culture and seed-bank verification (both specialists × 2 cycles). Full matrix in ConOps Slide 18.

Contingency triggers (numeric): Yellow when single-crop yield <85% of forecast OR fermentation pH drift >0.3; Orange when ≥2 modules show <70% nominal OR water recovery <95% for 3 consecutive sols OR ClO₄⁻ residual >0.020 wt%; Red on total module failure OR contamination event OR reserve burn-rate >2× nominal. Red triggers <24 h Earth comm and 100% reserve-ration switch.

Role	Startup (Sols 1–60)	Nominal (60–450)	Contingency
Food Systems Engineer	9 h/sol module restart	9 h/sol cultivation/HACCP	9 h/sol crisis ops + 3 h/sol backup surge
Nutrition / Meal Specialist	9 h/sol HACCP re-validation	9 h/sol meal prep + DGA review	9 h/sol ration logistics + 3 h/sol backup
4 Cross-trained backups	2 h/sol shadow training	2 h/sol on-call	8 h/sol redirected from EVA/science

HACCP critical control points (10): perchlorate residual, heavy-metal Pb/As, cyanotoxin screen, Tenebrio rearing temp, fermentation pH/Aw, probiotic count, listeria/salmonella swab, water Aw/microbial, ration shelf-life, allergen segregation.

Risk register and mitigation (R1–R8):

Risk	Pr/Sev	Mitigation
R1 · HybridSoil mass balance gap	L/L	Closed: 2.2% bio-polymer binder (lab-validated)
R2 · ClO ₄ ⁻ stoichiometry deviation	M/L	Phoenix MECA wet-chem replication; Sol-24 gate
R3 · DGA fitness without baseline / CV	L/M	Reported 92.2% with 5-fold CV ±0.42, 95% CI; quarterly re-eval
R4 · Anabaena N ₂ fix at 100 hPa pN ₂	M/M	CyanoGrow N redundancy via FermentLab; backup NH ₄ ⁺ stock
R5 · 34% O ₂ hyperoxia fire risk	M/H	Fire-rated enclosures, O ₂ trim, NASA-STD-3001 Vol.2 Sec. 6.4
R6 · NEXUS thresholds clinical validation	M/L	Calibrated vs. ISS + MARS500 baselines (HRV ms, cortisol nmol/L)

R7 · Probiotic viability in hyperoxia	M/M	Anaerobic 1.0 psi sealed cartridges; viability check per 14-sol cycle
R8 · Earth-provisioned 47.0% by cal (cap 50%) audit	L/L	Calorie budget table delivered (Sec. 4)

5b. Multi-Year Sustainability. ESPERANZA II supports 4+ consecutive 500-sol crew rotations (≈ 5.6 Earth years; Rules p.6 multi-year objective), with end-of-rotation hand-over (Sols 450–500: seed-bank verification, culture refresh, 50-sol transfer to incoming crew). Lifecycle: LED grow array $\geq 50,000$ h; hydroponic membranes 18–24 mo (replace each rotation); cyano starter cultures 12 mo (refresh per hand-over); seed bank ≥ 5 yr frozen (re-stock every 3 rotations). Multi-year mass balance: ≈ 91 M crew-kcal over 4 rotations (22.76 M kcal/rotation $\times 4$), $\approx 53\%$ met in-situ, no mid-turn resupply — see ConOps Multi-Year slide.

6. DATA FOUNDATION, NOVELTY AND TERRESTRIAL IMPACT

Three verified NASA datasets ground every result; no simulants. Mars regolith — NASA PDS (7 missions, 35 samples; XRFs, APXS, ChemCam LIBS, PIXL, MECA). Lunar regolith — JSC Lunar Sample Curator (47 samples, 6 Apollo missions + LRO). ISS microbiome — NASA Astronaut Microbiome Project, Twins Study and MARS500 (400 records; Voorhies 2019, Turrone 2020).

A 74-publication review (Duri 2022, Fackrell 2024, Verseux 2021) finds no prior system integrating lunar+Mars ISRU, perchlorate bioremediation, DGA-evolved probiotics, radiation-medicine fermentation and closed-loop biometric AI for a 500-sol crewed mission.

Terrestrial impact: MarsRem for arid-soil remediation (>500 M people); DGA-evolved probiotics for IBD (~ 10 M patients); CyanoGrow N_2 -fixation for depleted soils. All five pillars are dual-use.

7. KEY QUANTITATIVE RESULTS

Butyrate 87.9 mmol/sol ($5\times$ ISS); personalised 96.5/100. · 14-sol plan: 78 unique meals @ 3,035 kcal/sol + 200 kcal/EVA-h at 8.0 crew-EVA-h/sol nominal, produced in situ; ~ 123 g protein (18 g from 41 g/sol Tenebrio flour), ~ 58 g fibre; DRI-compliant; live DGA. · NASA-STD-3001 Vol.2 §7 (all 12 food-system attributes addressed), Vol.2 §6.4 & §4, Vol.1 §5; specialist crew-time 81.2/90 h per 5-sol (90%); 47.0% Earth-provisioned. Figure 2 shows the loop; parameters in ConOps Appendix A8.

8. Team and Affiliations. María Jesús Puerta Angulo is an independent Mining Engineer (Tarragona, Spain) specializing in closed-loop life support and ISRU, sole international winner of the 2025 NASA LunaRecycle Challenge (Phase 1). ESPERANZA II is a solo entry.

Intellectual Property. All concepts, algorithms and models are the sole intellectual property of María Jesús Puerta Angulo; no third-party proprietary technology is knowingly incorporated into the flight architecture. AI Disclosure (Rules pp. 7-8). Generative AI was used under human-in-the-loop supervision: Anthropic Claude assisted with drafting, editing and validation review of narrative text; ElevenLabs rendered the video narration from the author's own cloned voice — the only AI-generated asset in this entry, with no AI-generated imagery, figures or footage. No scientific claim, dataset, calculation or design decision was AI-generated; the author bears sole responsibility.

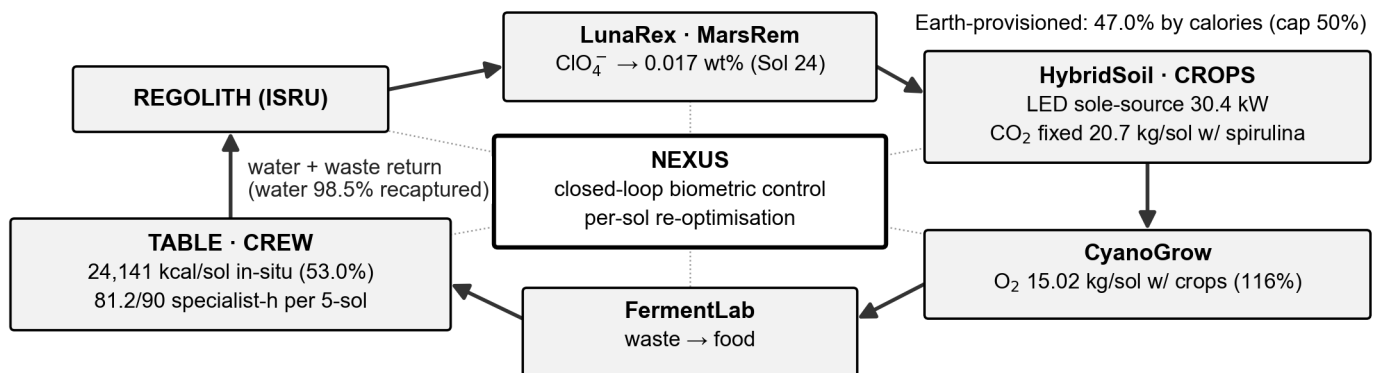


Figure 2. The ESPERANZA II closed loop — six stages, one control layer; nothing leaves the system.

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